

Tax Evasion and Productivity: Identifying and Correcting for Cost Overreporting*

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Abstract

Value-added taxes (VAT) allow firms to deduct the taxes paid on purchases from the taxes collected on sales, creating an incentive for firms to overreport the purchases of inputs to reduce their tax bill. The fundamental issue in detecting input overreporting stems from two unobservables, true inputs and productivity. Because firms differ in productivity, a firm reporting high input use, for a given level of output, may be either overreporting or simply less productive. I use a structural production-function approach and a benchmark group, a subset of firms that report inputs correctly, to detect and measure overreporting. Because the benchmark firms are assumed to share the same technology as the other firms, I can estimate its parameters. Using these estimates, I generate predictions and compare how potentially tax-evading firms would behave without overreporting to what they report. I apply the method to a Colombian manufacturing survey from 1981 to 1991. During this period, the Colombian sales tax worked as a VAT and, due to government and market scrutiny, a subset of firms reported their inputs correctly.

My approach yields a simple yet informative statistical test to detect input overreporting. Under the null hypothesis of correct reporting, the test is robust to a wide family of specifications of both the probability of detection and the cost of overreporting. Among the 5 largest industries, the test rejects the null hypothesis in all 4 industries subject to the sales tax, and, as expected, it fails to reject in the industry exempt from the sales tax. Across the 7 largest industries where the null is rejected, the median input overreporting, as a share of true inputs, amounts to 16 percent. In addition, I find that, between 1983 and 1987, after a fiscal reform raised the sales tax in Colombia, firms in affected industries increased the ratio of overreported to true inputs by about 8.5 percentage points, an implied median-based overreporting elasticity of the tax rate between 4.7 and 9.5.

The magnitude of detected overreporting matters for methods of estimating production functions that exploit a flexible input to identify productivity (Levinsohn and Petrin 2003; Doraszelski and Jaumandreu 2013; Akerberg et al. 2015; Gandhi et al. 2020). These methods typically assume that materials are flexible, and their demand or first-order conditions reveal productivity. However, materials are commonly tax deductible and, thus, the input firms have the

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incentive to overreport. I show that failing to correct for overreporting yields output elasticities of materials that are larger, and productivity distributions that are more dispersed and less persistent. Using the production-function parameters, I estimate a structural model of firm input overreporting to ask how does the expected claimed deductions, and thus government revenue, respond to the tax rate, where expected claimed deductions include true and undetected fictitious deductions. Approaching the observed tax rate from the left, the expected deductions elasticity of the tax rate is -9.7. Undetected overreporting costs the government around 3 percent of potential sales-tax revenue.

JEL Classification: H26, D22, L60, O47

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Introduction

Firms owned by the highest earners evade more tax through cost overreporting than firms owned by lower-income earners: relative to the bottom 80 percent of the income distribution, the tax evaded in Ecuador by the top 5 percent, as a share of owners' income, is 56 times larger (Carrillo et al. 2022). Despite its relevance, we still do not know much about firms' cost overreporting. Tax authorities around the world document this kind of tax evasion, often through fictitious invoices (OECD 2017), and the losses they report are significant: between 1.2 and 2.7 percent of total tax revenue in Chile, Colombia, Ecuador, Poland, and Slovakia, and about a ninth of the entire VAT compliance gap in Colombia and Slovakia (OECD 2017; Carrillo et al. 2022; CASE and IHS 2018; Longinotti 2022). Because some of these figures are tax administrations' own estimates, and all are conditional on detected evasion, true losses are plausibly larger.

Research has documented how individuals' reported income responds to tax rates, but has overlooked how firms' overreporting of inputs responds. For individuals, a large literature estimates how reported income responds to the tax rate and shows that the response includes evasion and avoidance as well as changes in real activity (Saez et al. 2012; Chetty 2009). For firms, evidence on responses to tax rates comes mostly from bunching at kinks and notches in the tax schedule (Best et al. 2015; Almunia and Lopez-Rodriguez 2018) and from misreported imports (Fisman and Wei 2004). Moreover, economic theory suggests that optimal commodity-tax rates should be lower on goods whose evasion is more elastic with respect to the tax rate (Cremer and Gahvari 1993). Yet, to my knowledge, no study has estimated how firms' overreporting of inputs, a form of tax evasion that the credit on purchases in value-added taxes incentivizes, responds to the tax rate. One reason is that true inputs are unobserved: for a given level of output, a firm reporting high input use may be overreporting, or it may suffer from low productivity.

I investigate whether input overreporting rises with the tax rate, and the extent to which it affects government tax revenue through firms' claimed deductions. Colombia's sales tax worked as a value-added tax, and in 1983 it raised the rate for most manufacturing industries but left a few exempt. I

find that, between 1983 and 1987, overreporting by unincorporated firms in the affected industries rose significantly, by about 8.5 percentage points of true materials. In a structural model estimated on all industries, I find that a 0.5 percent increase in the sales-tax rate raises claimed deductions by 16 to 28 percent, while cuts lower claimed deductions only gradually: the smallest cut distinguishable from current policy is 8 percent. A back-of-the-envelope calculation suggests that, as a share of potential net sales-tax revenue from unincorporated firms in the sample, undetected overreporting costs about 3 percent. Near the current rate, and holding the tax collected on sales fixed, the elasticity from the left of this revenue with respect to the tax rate is about -9 (**?@sec-app-backofenvelope**).

To distinguish between evasion and productivity, I employ a structural production-function approach, and then use it to study how evasion responds to the tax rate. In the absence of evasion, the first-order condition for materials, the flexible deductible input, reveals the structural relationship between the firm's materials share of revenue and the common technology within the industry. The government and market scrutiny keep corporations from overreporting. Corporations are thus the benchmark that allows me to identify the common technology in each industry and the distribution of measurement error in output. Among unincorporated firms, deviations from the common technology reveal overreporting up to measurement error in output. Deconvolution then separates the distribution of overreporting from that of the measurement error, and the production-function estimates recover productivity net of overreporting. To study how overreporting responds to the tax rate, I exploit the 1983 reform, measuring overreporting against the output elasticity of materials identified from corporations. Finally, employing an entropy-based method that does not require specifying the distribution of the unobservables (Schennach 2014), I estimate a structural model in which firms choose inputs and overreporting to maximize expected after-tax profits, weighing the tax evaded against the probability of detection and a cost of evasion that varies with productivity.

Separating overreporting from productivity also yields three findings of its own. First, a simple test compares unincorporated firms' materials shares with the corporate benchmark in each industry: under the null hypothesis of no overreporting, both should be equal; under the alternative, overreporting, the difference will be positive. Among the 20 largest industries, the test detects overreporting in 9 of the 18 industries liable for the sales tax, including 4 of the 5 largest, which together produce about two fifths of manufacturing output. It rejects decisively, at the 1 percent level, in 7 of them, where, in the median industry, unincorporated firms overreport an average of 16 percent of their true materials. It fails to reject in either of the 2 food industries, which were exempt from the sales tax and thus incentives to overreport were negligible. The test is valid for a broad class of models of detection and cost of evasion, and holds under any common production technology. A rejection is also unambiguous, because no firm gains by underreporting deductible inputs, so it can only point to overreporting. Second, failing to correct for overreporting biases estimates of the production function and of productivity. In those 7 industries, the uncorrected estimates (Gandhi et al. 2020) show output elasticities of materials that are consistently larger, and productivity distributions that are more dispersed and less persistent, than the corrected ones: in the median industry, the ratio

of the 90th to the 10th percentile is 66 percent larger and the half-life of productivity shocks is 52 percent shorter. Third, the structural estimates show a cost of evasion that is convex in productivity and lowest near the 86th percentile of productivity, which rationalizes a pattern observed in other studies: evasion rises with firm size but drastically drops at the very top (Carrillo et al. 2022).

These results relate to three strands of public economics and to one of productivity measurement. First, I estimate how a particular form of evasion by firms responds to the tax rate, a response that the literature on taxes and revenue has mostly studied through individuals' reported income. Traditionally, the elasticity of personal taxable income has combined all the different responses to the rate, evasion included, into a single statistic (Feldstein 1999; Slemrod and Kopczuk 2002; Saez et al. 2012). However, when part of the cost of evasion is a transfer rather than a resource cost, this elasticity alone no longer measures the efficiency cost of taxation (Chetty 2009). Measuring that cost requires knowing how evasion itself responds to the rate, separately from real activity. Direct evidence on this response is scarcer, and comes from the gap between households' consumption and reported income (Gorodnichenko et al. 2009) and, for firms, from bunching at kinks and notches in the tax schedule (Best et al. 2015; Almunia and Lopez-Rodriguez 2018). I provide evidence for firms' overreporting of inputs under a sales tax that credits purchases, from both a statutory rate change and a structural model that separates overreporting from productivity. The structural model also speaks to the theory of optimal commodity taxation under evasion by estimating how claimed deductions respond to the rate (Cremer and Gahvari 1993).

Second, I provide measures of firms' cost overreporting without relying on confidential tax records. Most evidence on firm evasion focuses on underreported revenue (Slemrod 2019), yet the cost side matters. Firms offset enforcement on revenue by raising reported costs (Carrillo et al. 2017; Asatryan and Peichl 2017), and paper trails make transactions between firms harder to misreport than sales to final consumers (Pomeranz 2015); therefore, firms that sell to other firms have little room to misreport on the sales side and more on the cost side. Cost overreporting has mostly been measured directly with transaction-level tax records in Ecuador and Mexico. These studies reveal the mechanism: firms acquiring fake invoices from ghost firms (Carrillo et al. 2022; Zumaya et al. 2021). Tax records, however, are confidential, available for few countries, and accessible only to selected researchers. Survey-based measures of firm evasion rely instead on firms' self-reported share of sales declared to the tax authority (Dabla-Norris et al. 2019), whose truthfulness cannot be verified. My approach needs only reported inputs and output, and is thus equally applicable to standard firm-level surveys, administrative records, and confidential tax records.

Third, I extend the validation-sample approach from individuals to firms. This approach infers evasion by comparing an indirect measure of true income between a truth-reporting group and a group that may misreport. For example, the self-employed report less income relative to their food consumption than employees, whose income is reported by their employers (Pissarides and Weber 1989). Researchers have also used the same comparison to measure how evasion responds to a tax

reform (Gorodnichenko et al. 2009). For firms, however, the natural indirect measure, output relative to inputs, also depends on an additional unobservable, productivity. A firm reporting high input use may be overreporting or may simply have a negative productivity shock. I use the production function to separate the two, with corporations as the truth-reporting group. Unlike the comparisons for individuals, my approach does not require the two groups to be alike in size or productivity: corporations are much larger than unincorporated firms, and the production function accounts for the differences in productivity behind this gap.

Finally, the paper shows how tax-motivated overreporting of inputs biases production-function estimates. Proxy-variable methods rely on a flexible input, typically materials, either inverting its demand (Levinsohn and Petrin 2003; Akerberg et al. 2015) or using its first-order condition (Gandhi et al. 2020). These methods treat reported inputs as true. Work on mismeasured inputs has focused on classical measurement error (mean zero and independent of the true input), mostly in capital (Collard-Wexler and De Loecker 2020). Overreporting is neither: firms choose it based on their own productivity. Furthermore, it affects exactly the input on which these methods rely. Failing to correct for overreporting yields output elasticities of materials that are consistently larger, and productivity distributions that are more dispersed and less persistent, than the corrected estimates. These findings matter for any analysis built on measured productivity differences across firms.

The rest of the paper follows the argument. **?@sec-model** shows how the tax rate generates incentives for a firm to overreport materials, and how its optimal choice depends on the probability of detection and the cost of evasion. **?@sec-setting** explains why Colombian corporations can serve as truth-tellers. **?@sec-testing** shows how overreporting can be detected and measured from production data alone. Then, **?@sec-pf** demonstrates how failing to correct for overreporting biases production-function and productivity estimates. **?@sec-fiscal** leverages the 1983 reform to document that overreporting responds to the tax rate, and **?@sec-counterfactual** turns the question around and asks how much revenue the government would lose to overreporting as it adjusts the tax rate. **?@sec-conclusion**, as tradition demands, concludes.

Online Appendix

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